MIPS: A Multimodal Infinite Polymer Sequence Pre-training Framework for Polymer Property Prediction

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Abstract

Polymers, composed of repeating structural units called monomers, are fundamental materials with a wide range of applications in daily life and industry. Accurate property prediction for polymers is essential for their design, development, and application. However, existing modeling approaches, which typically represent polymers by the constituent monomers, struggle to capture the whole properties of polymer, since the properties change during the polymerization process. In this study, we propose a Multimodal Infinite Polymer Sequence (MIPS) pre-training framework, which represents polymers as infinite sequences of monomers and integrates both topological and spatial information for comprehensive modeling. From the topological perspective, we generalize message passing mechanism (MPM) and graph attention mechanism (GAM) to infinite polymer sequences. For MPM, we demonstrate that applying MPM to infinite polymer sequences is equivalent to applying MPM on the induced star-linking graph of monomers. For GAM, we propose to further replace global graph attention with localized graph attention (LGA). Moreover, we show the robustness of the "star linking" strategy through an adversarial evaluation method named Repeat and Shift Invariance Test (RSIT). Despite its robustness, "star linking" strategy exhibits limitations when monomer side chains contain ring structures, a common characteristic of polymers, as it fails the Weisfeiler-Lehman (WL) test. To overcome this issue, we propose backbone embedding to enhance the capability of MPM and LGA on infinite polymer sequences. From the spatial perspective, we extract 3D descriptors of repeating monomers to capture spatial information. Finally, we design a cross-modal fusion mechanism to unify the topological and spatial information. Experimental validation across eight diverse polymer property prediction tasks reveals that MIPS achieves state-of-the-art performance. Ablation studies further comfirm the efficacy of our infinite polymer sequence modeling approach and multimodal pre-training framework.[1]

CCS Concepts

• **Applied computing → Bioinformatics**; • **Computing methodologies → Unsupervised learning**.

Keywords

Multimodal Pre-training, Polymer Property Prediction, Infinite Polymer Sequence Modeling

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1 Introduction

Polymers constitute a fundamental class of materials widely utilized in daily life [30] and industrial applications, such as energy storage [26], drug design [33], and optoelectronics [29], due to their versatile properties and functionalities. Precisely determining the properties of polymers is crucial for their design, development, and deployment, as it allows researchers to optimize performance for targeted applications. While conventional approaches to obtaining polymer properties, including experiments and simulations, provide accurate results, they typically remain time-consuming and resource-intensive. In contrast, machine learning approaches have emerged as a promising alternatives, significantly reducing time and cost, and therefore facilitating efficient screening of extensive polymer libraries for specific applications [21]. These approaches compute polymer descriptors as input features for traditional machine learning models. Inspired by the success of deep learning in natural language processing and computer vision [22], researchers apply neural networks to automatically learn features from raw

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[1]Our code is available at https://github.com/wjxts/MIPS

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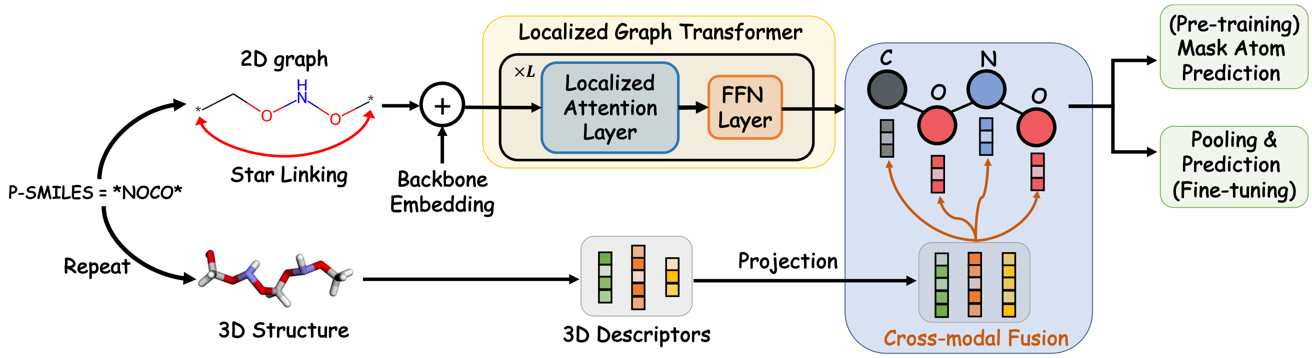

Figure 1: Pipeline of MIPS pre-training framework.

polymer data [34, 39]. In practice, the amount of labeled data are often limited. Self-supervised learning (SSL) methods have been further developed to leverage the unlabeled data which improeves the generalization ability [18, 44, 56]. Despite these advancements, given the inherent complexity of polymers due to structural variations and the impact of the polymerization process, developing effective and reliable modeling approaches remains a significant challenge. Existing polymer modeling methods primarily rely on monomer representations (P-SMILES[18]). However, due to the polymerization process, the properties of polymers differ from those of their constituent monomers [11, 31]. In this work, we present a Multimodal Infinite Polymer Sequence (MIPS) pre-training framework which models polymers as infinite sequences of monomers and effectively integrates both topological and spatial information via a cross-modal fusion mechanism. Our contributions can be summarized as follows:

- We propose to model polymers as infinite sequence of monomers, and accordingly extend message passing mechanism (MPM) and graph attention mechanism (GAM). We extend MPM to infinite polymer sequences by "star linking" strategy and further replace GAM with localized graph attention (LGA).
- We introduce Repeat and Shift Invariance Test (RSIT) to access the robustness of polymer modeling methods. Experimental results demonstrate that existing polymer modeling methods, including "star keep", "star remove", and "star substitution", suffer significant performance degradation under RSIT. In contrast, our "star linking" is resistant to RSIT.
- We theoretically prove that WL test, as well as the MPM and LGA, can not distinguish the defined twin polymer graphs. We propose backbone embedding to improve the expressive power of MPM and LGA in distinguishing twin polymer graphs.
- Our proposed MIPS framework effectively integrates topological and spatial polymer features. Extensive evaluation on eight polymer property prediction tasks demonstrate that MIPS outperforms existing methods, achieving state-of-the-art performance. Comprehensive ablation and interpretation analyses reveal that MIPS can learn meaningful polymer representations.

2 Related Work

2.1 Polymer Representation Learning

Acquiring effective polymer representation is essential for accurately predicting polymer properties [18, 50]. Classic quantitative structure-property relationship (QSPR) approaches [21] rely on handcrafted features, and utilize traditional machine learning models, such as support vector machines [8], random forests [6], principle component regression [14], etc. However, these techniques require extensive domain expertise for designing task-specific features, limiting their adaptability and generalization across different tasks. With the rise of deep learning, neural networks have been utilized to automatically learn features from raw polymer data. Leveraging polymer representations, such as P-SMILES [18], convolutional neural networks, recurrent neural networks and transformer models have been employed to establish effective mappings between polymer sequences and their corresponding properties. Despite their advances, the efficacy of these deep learning models largely depends on the availability of sufficiently large and labeled datasets, which are often scarce in practical scenarios. Motivated by the success of pretraining-finetuning paradiams in natural language processing [9] and contrastive learning strategies in computer vision [7], recent approaches have introduced masked-token prediction [18, 56] and contrastive methods [44] into polymer representation learning. Due to the importance of 3D structure information in deciding polymer property [40, 46], MMPolymer [44] further combine 3D information and 1D P-SMILES representation to enhance polymer property prediction. However, existing techniques predominantly rely on monomer-level P-SMILES, whose properties often differ from the resultant polymer structures after polymerization. To address these limitations and leverage multimodal information more effectively, we propose a Multimodal Infinite Polymer Sequence (MIPS) pre-training framework. MIPS represents polymers explicitly as infinite sequences of monomers and incorporates both topological and spatial information. Experimental results demonstrate that MIPS significantly outperforms existing methods across eight polymer property prediction tasks, setting new the state-of-the-art performance. Furthermore, ablation and interpretation studies reveal that MIPS can learn meaningful polymer representations.

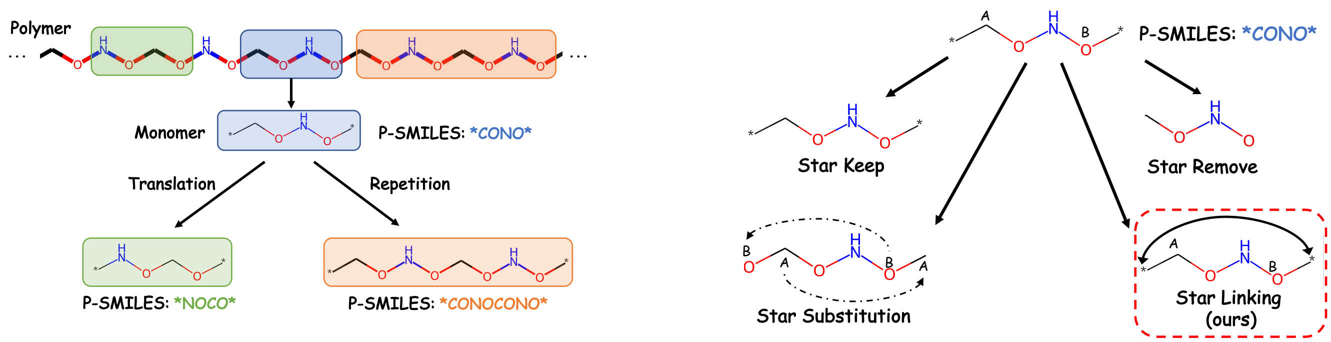

Figure 2: Left: How polymers are represented by P-SMILES, along with examples of translation and repetition. The underlying polymer of P-SMILES is unchanged under translation and repetition. Right: Three existing polymer modeling methods: "star keep", "star remove", and "star substitution". Our proposed "star linking" strategy is specifically designed to model infinite polymer sequences.

2.2 Molecular Self-supervised Learning

Existing polymer representation learning methods primarily rely on monomer-level P-SMILES [18], which allows monomers to be treated as special molecular instances. Under this perspective, a variety of molecular self-supervised learning (SSL) methods can be seamlessly adapted to polymer domain [56]. SSL leverages large volumes of unlabeled data to learn powerful feature representations. Molecular SSL methods can be broadly categorized according to the chosen molecular input form: string-based, graph-based, 3D-based, and hybrid approaches. String-based methods, such as MolBERT [10] and MolGPT [4], regard molecules as sequences of characters by SMILES [48] and utilize transformer architectures with mask-based token prediction to capture molecular representation. Graph-based methods incorporate graph-level SSL strategies, such as masked atom/bond prediction [35] and graph contrastive learning based on molcular augmentations [47]. Recognizing the importance of spatial information, 3D-based methods have been developed with denoising autoencoders [54] and equivariant neural networks [36] specifically designed to learn robust 3D structural representations. Additionally, since complementary information is contained within 2D topology and the 3D geometry, multiple hybrid methods have emerged [24, 40, 57] that combine both 2D and 3D representations for richer feature extraction. In this work, we extend the molecular SSL concepts to polymer representation learning. Our proposed MIPS framework represents polymers as infinite monomer sequences and simultaneously leverages topological and spatial information through a cross-modal fusion mechanism.

2.3 Expressive Power of Graph Neural Networks

The analysis of the expressive power of graph neural networks (GNNs) represents an active and significant area of research within theoretical graph learning [55]. Such analysis deepens our understanding of GNN capabilities and provides valuable insights and inspiration for developing more powerful architectures [51]. The seminal work of Xu et al. [51] establishs that the expressivity of GNNs in distinguishing non-isomorphic graphs is limited by the Weisfeiler–Lehman (WL) test, and further introduced the Graph Isomorphism Network (GIN), which achieves this theoretical upper bound. Subsequent studies have designed GNN variants capable of surpassing the WL test, thus achieving higher expressive power. For instance, K-GNN [28] enhances GNN expressivity by generalizing to higher-order architectures based on the multidimensional k-WL algorithm. $\mathcal{O}$-GNN [58], GSN-v [5], FragNet [49] explicitly incorporate the fragment bias into GNNs to improve representational capacity. In this work, we extend the previous analysis of GNN expressivity to infinite polymer graph sequences. We demonstrate that the standard WL test, as well as the message passing and localized graph attention mechanisms, can not distinguish between the defined twin polymer graphs. To address this limitation, we propose backbone embedding that enhances the expressive power of message passing and localized graph attention mechanisms for distinguishing twin polymer graphs. Experimental results confirm the effectiveness of the backbone embedding, especially in scenarios involving polymers that contain multiple ring structures.

2.4 Adversarial Attack

Although deep learning has machieved remarkable progress across various fields [22], neural networks are vulnerable to small perturbations in input data [41]. Such adversarial samples share semantic content identical to the original inputs, yet significantly degrade the performance of models, raising serious concerns about the reliability and robustness of deep learning approaches. In this work, we propose a Repeat and Shift Invariance Test (RSIT) to evaluate the robustness of polymer modeling methods. RSIT generates adversarial samples by randomly augmenting the polymer P-SMILES through sequence translation and repetition, without changing the underlying polymer. We show that existing polymer modeling methods, including "star keep", "star remove", and "star substitution" [44], exhibit significant performance degradation under RSIT. We model the invariance of translation and repetition by directly modeling infinite polymer sequences. Our infinite polymer sequence modeling approach is resistant to RSIT and achieve superior empirical performance, underscoring its robustness and effectiveness.

Algorithm 1: Random Augment P-SMILES

Input: P-SMILES s
Output: Augmented P-SMILES $\hat{s}$ = random_augment (s)

$\hat{s} \leftarrow s$;
$p \leftarrow$ `Bernoulli(0.5)`;
if $p == 1$ **then**
  $\hat{s} \leftarrow$ `repeat(`$\hat{s}$`); // e.g., *CONO* ⟶ *CONOCONO*`
$\hat{s} \leftarrow$ `random_translation(`$\hat{s}$`); // e.g., *CONO* ⟶ *NOCO*`
return $\hat{s}$

Algorithm 2: Repeat and Shift Invariance Test (RSIT)

Input: model m, number of trials T, sample x, label y, loss function L
Output: adversarial loss `adv_loss`, adversarial prediction `adv_predict`

for $i = 1$ *to* T **do**
  `adv_x` $\leftarrow$ random_augment(x) (Algorithm 1);
  $\hat{y} \leftarrow m.predict($`adv_x`$)$;
  `loss` $\leftarrow L(\hat{y}, y)$;
  if `loss > adv_loss` **then**
    `adv_loss` $\leftarrow$ `loss`;
    `adv_predict` $\leftarrow \hat{y}$;
return `adv_loss`, `adv_predict`

3 Methods

In this section, we introduce the Multimodal Infinite Polymer Sequence (MIPS) pre-training framework which models polymers as infinite sequences of monomers and integrates both topological and spatial information. The pipeline of MIPS is illustrated in Figure 1. MIPS consists of three parts: topological structure encoder, spatial structure encoder, and cross-modal fusion.

3.1 Topological Structure Encoder

Polymer sequences, composed of repeating units called monomers, are usually represented by P-SMILES [18], which encode the topological structures of monomers and two endpoints involved in polymerization. We show an illustration of P-SMILES in Figure 2 (left). The star symbols in P-SMILES indicate the endpoints of the monomer. *An important property of P-SMILES is that the polymer represented by P-SMILES remains invariant under translation and repetition of P-SMILES (shown in Figure 2 (left)).* To model such invariance, we propose to directly build model on infinite polymer sequences since infinite polymer sequences are naturally invariant under the translation and repetition of P-SMILES.

3.1.1 Repeat and Shift Invariance Test. To test how well the model can capture the translation and repetition invariance of P-SMILES, we propose Repeat and Shift Invariance Test (RSIT) (Algorithm 2). RSIT is an adversarial attack algorithm that generates adversarial samples by randomly augmenting the input P-SMILES through translation and repetition. We will show in the experiment section that existing modeling methods, including "star keep", "star remove", and "star substitution" (Figure 2 (right)) exhibit significant performance degradation under RSIT, while our infinite polymer sequence modeling approach is resistant to RSIT.

3.1.2 Modeling Infinite Polymer Sequences. In this subsection, we introduce how to model infinite polymer sequences using message passing mechanism (MPM) and graph attention mechanism (GTM). We define the monomer graph as $G = (V, E, \mathbf{X})$, where $V = \{v_i\}_{i=0}^{|V|-1}$ is the list of atoms in the monomer, $E = \{e_i\}_{i=1}^{|E|}$ is the list of bonds, $e_i = (v_{i1}, v_{i2})$, and $\mathbf{X} \in \mathbb{R}^{d\times|V|}$ is the atom feature matrix. Without loss of generality, we assume v_0 and $v_{|V|-1}$ are the boundary nodes of the monomer to form polymer. The polymer graph $G^p = (V^p, E^p, \mathbf{X}^p)$ is the periodic repetition of the monomer graph G, where the i-th ($i \in \mathbb{Z}$) atom of V^p has the same properties as the (i *mod* $|V|$)-th atom of V. Except for the bonds in each monomer, E^p also contains the bonds between the boundary atoms of adjacent monomers, i.e., $v^p_{k|V|}$ and $v^p_{k|V|-1}$, $\forall\, k \in \mathbb{Z}$.

Message Passing Mechanism on Infinite Polymer Sequences. Denote the neighborhood of node v as $\mathcal{N}(v_i)$, and the node feature of v as $\mathbf{x}_v$. The message passing mechanism [12, 16, 43, 51] updates the node feature of v as follows:

$$\mathbf{x}_v \leftarrow \text{UPDATE}\left(\mathbf{x}_v, \text{AGG}\left(\{\mathbf{x}_u\}_{u\in\mathcal{N}(v)}\right)\right), \tag{1}$$

where UPDATE and AGG serve as the update and message aggregation functions, respectively. Typical networks using message passing mechanism include Graph Convolutional Network (GCN) [16], Graph Isomorphism Network (GIN) [51], Graph Attention Network (GAT) [43], GraphSAGE [12], etc. Directly applying message passing mechanism on polymer graph is prohibitive due to the infinite number of nodes in the polymer graph. However, we will show that message passing on infinite polymer sequences is equivalent to message passing on the induced star-linking graph of monomers. All the proofs are provided in the appendix.

Definition 1 (Induced Star-Linking Graph G^*)**.** The induced star linking graph $G^* = (V^*, E^*, \mathbf{X}^*)$ of the monomer graph $G = (V, E, \mathbf{X})$ is constructed by linking the boundary atoms of G, where $V^* = V$, $E^* = E \cup \{\left(v_0, v_{|V|-1}\right)\}$, $\mathbf{X}^* = \mathbf{X}$.

We assume G, G^p, and G^* share the same initial node features, i.e., $\mathbf{x}_i = \mathbf{x}^p_i = \mathbf{x}^*_i$, $\forall 0 \le i < |V|$.

Proposition 1. *Under the same message passing mechanism (Equation* (1)*), the node feature of the i-th atom in the polymer graph G^p is the same as the node feature of the i-th atom in the induced star linking graph G^*, $\forall 0 \le i < |V|$.*

Theorem 1. *If the network is composed of message passing layers (Equation* (1)*), nodewise transformations, and end with a mean pooling, Then the output of the network on the polymer graph G^p is the same as that on the induced star linking graph G^*.*

Using Theorem 1, we can directly apply the message passing mechanism on the induced star-linking graph G^* to effectively model infinite polymer sequences.

Localized Graph Attention Mechanism on Infinite Polymer Sequences. To improve the model capability and capture the long range dependency on graphs, graph attention mechanism is introduced [38, 42, 53]. Given the graph $G = (V, E, \mathbf{X})$, the attention mechanism computes $\mathbf{Y} = \text{Attn}(V, E, \mathbf{X})$ as follows:

$$\mathbf{Q} = \mathbf{W}^Q\mathbf{X},\ \mathbf{K} = \mathbf{W}^K\mathbf{X},\ \mathbf{V} = \mathbf{W}^V\mathbf{X},$$
$$\mathbf{A} = \text{Softmax}\left(\frac{\mathbf{K}^T\mathbf{Q}}{\sqrt{d}} + \mathbf{A}^d + \mathbf{A}^p\right),$$

$\mathbf{W}^Q$, $\mathbf{W}^K$ and $\mathbf{W}^V \in \mathbb{R}^{d\times d}$ are the projection weight matrices for query ($\mathbf{Q}$), key ($\mathbf{K}$) and value ($\mathbf{V}$) matrices, respectively. The softmax operation is applied column-wise to $\mathbf{A}$. $\mathbf{A}^d$ and $\mathbf{A}^p$ are distance attention and path attention. Assume the distance between node i and node j in graph is d_{ij}, and the shortest path between node i and node j is $\mathbf{p}_{ij} = (i_0, \ldots, i_{d_{ij}})$, then $\mathbf{A}^d$ and $\mathbf{A}^p$ are computed as:

$$A^d_{ij} = f_{\text{dist}}(d_{ij})$$
$$A^p_{ij} = f_{\text{path}}(\mathbf{p}_{ij}),$$

where f_{dist} and f_{path} are functions with learnable parameters and output scalar.

$$\mathbf{Y} = \mathbf{VA} \tag{2}$$

Directly applying global graph attention mechanism to the polymer graph is infeasible due to its infinite number of nodes. To overcome the infinite receptive field issue, we propose to replace the global graph attention mechanism with localized graph attention (LGA). LGA computes $\hat{\mathbf{Y}} = \text{LocalAttn}(V, E, \mathbf{X})$ by substituting Equation (2) with the following equation:

$$\begin{aligned} \hat{\mathbf{A}}_{ij} &= \mathbf{A}_{ij} * \mathbf{1}_{\{d_{ij}<d_{thres}\}} \\ \hat{\mathbf{Y}} &= \mathbf{V}\hat{\mathbf{A}}. \end{aligned} \tag{3}$$

$\mathbf{1}_{\{d_{ij}\le d_{thres}\}}$ equals to 1 if the distance between node i and node j is less than d_{thres}, otherwise 0.

Similar to message passing mechanism, we will show that localized graph attention on infinite polymer sequences is equivalent to localized graph attention on monomer with star linking strategy.

Theorem 2. *Assume that the distance between boundary atoms in monomer graph G is larger than $2d_{thres} - 1$. If the network consists of localized graph attention layers, nodewise transformations, and concludes with a mean pooling, then its output on the polymer graph G^p will be identical to that on the induced star-linking graph G^*.*

If the distance between boundary atoms in monomer graph G is less than $2d_{thres} - 1$, we first repeat the monomer graph G until the distance exceeds $2d_{thres} - 1$, Then, we apply the localized graph attention mechanism to the augmented monomer graph.

3.1.3 Backbone Embedding.

Although message passing mechanism is powerful, capable of performing graph isomorphism tests with the same expressive power as the WL test [51], we will demonstrate that it often fails in the case of infinite polymer sequences with ring structures which are ubiquitous in practice.

Definition 2 (Twin Polymer Graphs). Assume the monomer graphs of two polymers G^p_1 and G^p_2 are G_1 and G_2, respectively, with $G^p_1 \neq G^p_2$. If their induced star-linking graphs G^*_1 and G^*_2 satisfies $G^*_1 = G^*_2$, we call $\left(G^p_1, G^p_2\right)$ a pair of twin polymer graphs.

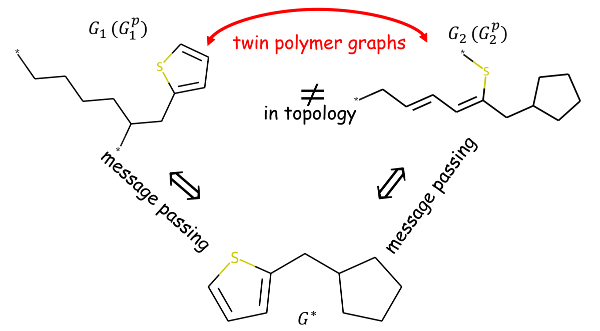

Figure 3: An example of twin polymer graphs $\left(G^p_1, G^p_2\right)$. By Theorem 1, message passing mechanism on G^p_1/G^p_2 is equivalent to that on G^*.

We show an example of twin polymer graphs in Figure 3.

Lemma 1. *WL test can not distinguish twin polymer graphs $\left(G^p_1, G^p_2\right)$.*

Theorem 3. *On twin polymer graphs $\left(G^p_1, G^p_2\right)$, massage passing mechanism (MPM, Equation (1)) or localized graph attention (LGA, Equation (3)) will produce identical outputs. In other words, MPM and LGA are unable to distinguish twin polymer graphs.*

To address the issue of message passing and localized attention mechanisms on infinite polymer sequences with ring structures, we propose *backbone embedding* to improve the capability of message passing and localized attention mechanisms. We define the backbone of monomer graph as the atoms and rings on the shortest path between the boundary atoms. We add a learnable embedding, named backbone embedding, to the backbone atoms to distinguish the backbone and the rings on the side chain. The backbone bone embedding $\mathbf{b}_i$ of node i is defined as:

$$\mathbf{b}_i = \begin{cases} \mathbf{b} & \text{if } i \text{ is a backbone atom} \\ 0 & \text{otherwise} \end{cases},$$

where $\mathbf{b} \in \mathbb{R}^d$ is the learnable vector.

3.1.4 Localized Graph Transformer.

Assume we are given the induced star-linking graph $G^* = (V^*, E^*, \mathbf{X}^*)$. We construct the backbone embedding matrix $\mathbf{B} \in \mathbb{R}^{d\times|V|}$ by stacking the backbone embedding of all atoms. Next, we initialize the node representation for the graph as $\mathbf{X}^t_0 = \mathbf{X}^* + \mathbf{B}$. Localized graph transformer consists of L localized graph attention layers. For the l-th layer, the node feature is updated as follows:

$$\mathbf{X}^t_{l+\frac{1}{2}} = \text{LayerNorm}\left(\text{LocalAttn}\left(V^*, E^*, \mathbf{X}^t_l\right) + \mathbf{X}^t_l\right),$$

$$\mathbf{X}^t_{l+1} = \text{LayerNorm}\left(\text{FFN}(\mathbf{X}^t_{l+\frac{1}{2}}) + \mathbf{X}^t_{l+\frac{1}{2}}\right),$$

where FFN is a two-layer MLP with ReLU activation [1]. We adopt the same distance attention and path attention mechanisms as proposed in Ying et al. [53].

Table 1: R^2 Performance of GIN3-512 with "star keep", "star remove", "star substitution", and our proposed "star linking" strategies w. and w.o RSIT with $T = 5$ trials. RSIT Gap: Average performance drop under RSIT.

	Egc	Egb	Eea	Ei	Xc	EPS	Nc	Eat	RSIT Gap
Star Keep	$0.866_{\pm 0.013}$	$0.901_{\pm 0.012}$	$0.890_{\pm 0.033}$	$0.762_{\pm 0.062}$	$0.252_{\pm 0.090}$	$0.723_{\pm 0.052}$	$0.814_{\pm 0.034}$	$0.956_{\pm 0.005}$	–
Star Remove	$0.857_{\pm 0.015}$	$0.881_{\pm 0.013}$	$0.871_{\pm 0.044}$	$0.772_{\pm 0.054}$	$0.243_{\pm 0.116}$	$0.714_{\pm 0.051}$	$0.805_{\pm 0.033}$	$0.939_{\pm 0.008}$	–
Star Substitution	$0.868_{\pm 0.011}$	$0.916_{\pm 0.012}$	$0.904_{\pm 0.035}$	$0.779_{\pm 0.056}$	$0.268_{\pm 0.108}$	$\mathbf{0.773}_{\pm 0.038}$	$0.845_{\pm 0.038}$	$0.963_{\pm 0.003}$	–
Star Linking (ours)	$\mathbf{0.870}_{\pm 0.010}$	$\mathbf{0.918}_{\pm 0.010}$	$\mathbf{0.916}_{\pm 0.025}$	$\mathbf{0.785}_{\pm 0.060}$	$\mathbf{0.279}_{\pm 0.130}$	$0.772_{\pm 0.048}$	$\mathbf{0.847}_{\pm 0.037}$	$\mathbf{0.966}_{\pm 0.006}$	–
Star Keep + RSIT	$0.745_{\pm 0.020}$	$0.780_{\pm 0.017}$	$0.593_{\pm 0.110}$	$0.524_{\pm 0.154}$	$-0.287_{\pm 0.271}$	$0.466_{\pm 0.089}$	$0.627_{\pm 0.069}$	$0.928_{\pm 0.012}$	0.233
Star Remove + RSIT	$0.765_{\pm 0.026}$	$0.792_{\pm 0.035}$	$0.696_{\pm 0.055}$	$0.468_{\pm 0.207}$	$-0.181_{\pm 0.291}$	$0.468_{\pm 0.085}$	$0.615_{\pm 0.073}$	$0.864_{\pm 0.016}$	0.199
Star Substitution + RSIT	$0.796_{\pm 0.020}$	$0.871_{\pm 0.020}$	$0.821_{\pm 0.043}$	$0.645_{\pm 0.081}$	$0.041_{\pm 0.142}$	$0.638_{\pm 0.074}$	$0.729_{\pm 0.056}$	$0.922_{\pm 0.012}$	0.107
Star Linking (ours) + RSIT	$\mathbf{0.870}_{\pm 0.010}$	$\mathbf{0.918}_{\pm 0.010}$	$\mathbf{0.916}_{\pm 0.025}$	$\mathbf{0.785}_{\pm 0.060}$	$\mathbf{0.279}_{\pm 0.130}$	$\mathbf{0.772}_{\pm 0.048}$	$\mathbf{0.847}_{\pm 0.037}$	$\mathbf{0.966}_{\pm 0.006}$	**0.000**

Table 2: R^2 Performance of GIN3-512 w./w.o backbone embedding (BE).

	Egc	Egb	Eea	Ei	Xc	EPS	Nc	Eat
Star Linking	$0.870_{\pm 0.010}$	$0.918_{\pm 0.010}$	$0.916_{\pm 0.025}$	$\mathbf{0.785}_{\pm 0.060}$	$0.279_{\pm 0.130}$	$\mathbf{0.772}_{\pm 0.048}$	$0.847_{\pm 0.037}$	$0.966_{\pm 0.006}$
Star Linking + BE	$\mathbf{0.882}_{\pm 0.007}$	$\mathbf{0.920}_{\pm 0.010}$	$\mathbf{0.917}_{\pm 0.029}$	$\mathbf{0.785}_{\pm 0.062}$	$\mathbf{0.346}_{\pm 0.083}$	$\mathbf{0.772}_{\pm 0.053}$	$\mathbf{0.849}_{\pm 0.043}$	$\mathbf{0.967}_{\pm 0.004}$

3.2 Spatial Structure Encoder

We extract the spatial information of polymers by computing the 3D descriptors of monomers. 3D descriptors are computed by the 3D coordinates of atoms in the monomer. We use the 3D molecular descriptors in Yang et al. [52], and atom pair 3D descriptors [3, 20]. Denote the 3D descriptors of the polymer as $\{\mathbf{x}_i^s\}_{i=1}^{N_s}$, $\mathbf{x}_i^s \in \mathbb{R}^{d_i}$. We use a linear projection to encode the 3D descriptors to the same dimension as the topological structure encoder:

$$\hat{\mathbf{x}}_i^s = \mathbf{W}_i \mathbf{x}_i^s,\ \forall\ i \in \{1, \ldots, N_s\},$$

where $\mathbf{W}_i \in \mathbb{R}^{d \times d_i}$ is the linear projection matrix. We stack $\hat{\mathbf{x}}_i^s$ as the spatial structure feature matrix $\mathbf{X}^s \in \mathbb{R}^{d \times N_s}$.

3.3 Cross-modal Fusion

With the topological structure feature matrix $X^t = X_L^t$ and the spatial structure feature matrix X^s, we propose a cross-modal fusion mechanism to fuse the topological and spatial information of polymers. To this end, we use a cross-attention layer [42] to integrate the spatial information to each atom in the topological structure space. $\mathbf{X}^{ts} = \text{CrossModalFusion}(X^t, X^s)$ is computed as follows:

$$\mathbf{Q} = \mathbf{W}^Q \mathbf{X}^t,\ \mathbf{K} = \mathbf{W}^K \mathbf{X}^s,\ \mathbf{V}^s = \mathbf{W}^V \mathbf{X}^s,$$

$$\mathbf{A}^{ts} = \text{Softmax}\left(\frac{\mathbf{K}^{\mathrm{T}}\mathbf{Q}}{\sqrt{d}}\right),$$

$$\mathbf{X}^{ts} = \text{LayerNorm}\left(\mathbf{X}^t + \mathbf{V}^{\mathrm{s}}\mathbf{A}^{ts}\right).$$

3.4 Pre-training Objective

We adopt masked-atom prediction as the pre-training objective [23, 35, 45]. Specifically, we mask the complete features of atoms within the polymer graph with a probability of p_{mask}. The type of the masked atoms is then predicted using the outputs of the cross-modal fusion layer, which are processed by a linear classifier.

Table 3: Brief statistics and descriptions of downstream datasets. E: Electronic, TP: Thermodynamic and Physical, OD: Optical and Dielectric.

Dataset	Property	Task Type	# of Samples	Unit	Data Range
Egc	bandgap (chain)	E	3380	eV	[0.02, 8.30]
Egb	bandgap (bulk)	E	561	eV	[0.39, 10.05]
Eea	electron affinity	E	368	eV	[0.39, 4.61]
Ei	ionization energy	E	370	eV	[3.55, 9.61]
Eat	atomization energy	TP	390	eV·atom^{-1}	[6.83, 5.02]
Xc	crystallization tendency	TP	432	%	[0.13, 98.41]
EPS	dielectric constant	OD	382	1	[2.61, 8.52]
Nc	refractive index	OD	382	1	[1.48, 2.58]

4 Experiments and Results

Dataset. Following Wang et al. [44], we utilize PL1M [27] as the dataset for multimodal pre-training, which contains approximately 1 million unlabeled polymer sequences. For the downstream tasks, we employ eight polymer property prediction datasets introduced in Künneth et al. [19], Zhang et al. [56], derived from the density functional theory calculations [32]. A brief summary of downstream datasets is provided in Table 3. These datasets cover a wide variety of polymer properties, enabling a comprehensive evaluation of our model's performance. We employ root mean square error (RMSE) or R-squared (R^2) over the five folds in cross-validation as the evaluation metrics. These metrics align with those used in previous studies [44, 56], ensuring a fair comparison.

4.1 Preliminary Study

We first conduct a preliminary study to demonstrate the effectiveness and robustness of our infinite polymer sequence modeling approach. For this purpose, we train a three-layer GIN model [51] with "start keep", "star remove", "star substitution", and our proposed "star linking" strategies on eight polymer property prediction tasks. The model, referred to as GIN3-512, is configured with 512 hidden units. We use the same atom features $\mathbf{X}^*$ as in Li et al. [23]

Table 4: The performance comparison of different methods on eight polymer property datasets, and the best result for each polymer property dataset has been bolded.

Metric	Method	Egc	Egb	Eea	Ei	Xc	EPS	Nc	Eat
RMSE (↓)	ChemBERTa [2]	$0.539_{\pm 0.049}$	$0.664_{\pm 0.079}$	$0.350_{\pm 0.036}$	$0.485_{\pm 0.086}$	$18.711_{\pm 1.396}$	$0.603_{\pm 0.083}$	$0.140_{\pm 0.010}$	$0.219_{\pm 0.056}$
	MolCLR [47]	$0.587_{\pm 0.024}$	$0.644_{\pm 0.072}$	$0.404_{\pm 0.017}$	$0.533_{\pm 0.053}$	$21.719_{\pm 1.631}$	$0.631_{\pm 0.045}$	$0.117_{\pm 0.015}$	$0.094_{\pm 0.033}$
	3D Infomax [40]	$0.494_{\pm 0.039}$	$0.553_{\pm 0.032}$	$0.335_{\pm 0.055}$	$0.449_{\pm 0.086}$	$19.483_{\pm 2.491}$	$0.582_{\pm 0.054}$	$0.101_{\pm 0.018}$	$0.094_{\pm 0.039}$
	Uni-Mol [57]	$0.489_{\pm 0.028}$	$0.531_{\pm 0.055}$	$0.332_{\pm 0.027}$	$0.407_{\pm 0.080}$	$17.414_{\pm 1.581}$	$0.536_{\pm 0.053}$	$0.095_{\pm 0.013}$	$0.084_{\pm 0.034}$
	SML [56]	$0.489_{\pm 0.056}$	$0.547_{\pm 0.110}$	$0.313_{\pm 0.016}$	$0.432_{\pm 0.060}$	$18.981_{\pm 1.258}$	$0.576_{\pm 0.020}$	$0.102_{\pm 0.010}$	$0.062_{\pm 0.014}$
	PLM [56]	$0.459_{\pm 0.036}$	$0.528_{\pm 0.081}$	$0.322_{\pm 0.037}$	$0.444_{\pm 0.062}$	$19.181_{\pm 1.308}$	$0.576_{\pm 0.060}$	$0.100_{\pm 0.010}$	$0.050_{\pm 0.010}$
	polyBERT [18]	$0.553_{\pm 0.011}$	$0.759_{\pm 0.042}$	$0.363_{\pm 0.037}$	$0.526_{\pm 0.068}$	$18.437_{\pm 0.560}$	$0.618_{\pm 0.049}$	$0.113_{\pm 0.003}$	$0.172_{\pm 0.016}$
	Transpolymer [50]	$0.453_{\pm 0.007}$	$0.576_{\pm 0.021}$	$0.326_{\pm 0.040}$	$0.397_{\pm 0.061}$	$17.740_{\pm 0.732}$	$0.547_{\pm 0.051}$	$0.096_{\pm 0.016}$	$0.147_{\pm 0.093}$
	MMPolymer [44]	$0.431_{\pm 0.017}$	$0.496_{\pm 0.031}$	$0.286_{\pm 0.029}$	$0.390_{\pm 0.057}$	$16.814_{\pm 0.867}$	$0.511_{\pm 0.035}$	$0.087_{\pm 0.010}$	$0.061_{\pm 0.016}$
	MIPS (ours)	$\mathbf{0.429}_{\pm 0.014}$	$\mathbf{0.460}_{\pm 0.037}$	$\mathbf{0.267}_{\pm 0.018}$	$\mathbf{0.384}_{\pm 0.048}$	$\mathbf{16.435}_{\pm 0.576}$	$\mathbf{0.484}_{\pm 0.049}$	$\mathbf{0.083}_{\pm 0.010}$	$\mathbf{0.038}_{\pm 0.006}$
R^2 (↑)	ChemBERTa [2]	$0.880_{\pm 0.023}$	$0.881_{\pm 0.028}$	$0.888_{\pm 0.035}$	$0.745_{\pm 0.102}$	$0.365_{\pm 0.098}$	$0.682_{\pm 0.123}$	$0.643_{\pm 0.076}$	$0.590_{\pm 0.078}$
	MolCLR [47]	$0.858_{\pm 0.010}$	$0.882_{\pm 0.027}$	$0.854_{\pm 0.038}$	$0.689_{\pm 0.037}$	$0.176_{\pm 0.026}$	$0.683_{\pm 0.020}$	$0.764_{\pm 0.037}$	$0.885_{\pm 0.104}$
	3D Infomax [40]	$0.900_{\pm 0.016}$	$0.898_{\pm 0.018}$	$0.891_{\pm 0.045}$	$0.766_{\pm 0.086}$	$0.274_{\pm 0.122}$	$0.690_{\pm 0.063}$	$0.797_{\pm 0.086}$	$0.869_{\pm 0.097}$
	Uni-Mol [57]	$0.901_{\pm 0.013}$	$0.925_{\pm 0.011}$	$0.901_{\pm 0.027}$	$0.820_{\pm 0.075}$	$0.454_{\pm 0.079}$	$0.751_{\pm 0.085}$	$0.828_{\pm 0.072}$	$0.937_{\pm 0.032}$
	SML [56]	$0.901_{\pm 0.022}$	$0.920_{\pm 0.029}$	$0.915_{\pm 0.015}$	$0.802_{\pm 0.051}$	$0.340_{\pm 0.125}$	$0.726_{\pm 0.038}$	$0.812_{\pm 0.058}$	$0.967_{\pm 0.015}$
	PLM [56]	$0.911_{\pm 0.014}$	$0.925_{\pm 0.021}$	$0.910_{\pm 0.019}$	$0.791_{\pm 0.049}$	$0.330_{\pm 0.105}$	$0.726_{\pm 0.058}$	$0.817_{\pm 0.056}$	$0.980_{\pm 0.008}$
	polyBERT [18]	$0.875_{\pm 0.006}$	$0.844_{\pm 0.034}$	$0.880_{\pm 0.035}$	$0.705_{\pm 0.085}$	$0.384_{\pm 0.066}$	$0.681_{\pm 0.058}$	$0.769_{\pm 0.034}$	$0.672_{\pm 0.119}$
	Transpolymer [50]	$0.916_{\pm 0.002}$	$0.911_{\pm 0.008}$	$0.902_{\pm 0.036}$	$0.830_{\pm 0.059}$	$0.430_{\pm 0.058}$	$0.744_{\pm 0.075}$	$0.826_{\pm 0.071}$	$0.800_{\pm 0.172}$
	MMPolymer [44]	$0.924_{\pm 0.006}$	$0.934_{\pm 0.008}$	$0.925_{\pm 0.025}$	$0.836_{\pm 0.053}$	$0.488_{\pm 0.072}$	$0.779_{\pm 0.052}$	$0.864_{\pm 0.036}$	$0.961_{\pm 0.018}$
	MIPS (ours)	$\mathbf{0.926}_{\pm 0.006}$	$\mathbf{0.945}_{\pm 0.007}$	$\mathbf{0.940}_{\pm 0.018}$	$\mathbf{0.846}_{\pm 0.051}$	$\mathbf{0.506}_{\pm 0.074}$	$\mathbf{0.814}_{\pm 0.045}$	$\mathbf{0.877}_{\pm 0.046}$	$\mathbf{0.990}_{\pm 0.004}$

to train GIN3-512 for 50 epochs with a batch size of 32 and learning rate of 0.001. Additionally, we evaluate the robustness of these strategie under RSIT, with results summarized in Table 1. We find that infinite polymer sequence modeling approach with "star linking" strategy outperforms the other three strategies on seven our of eight tasks without RSIT. Under RSIT, the performance of the three alternative strategies suffers significant degradation, while "star linking" strategy is resistant to RSIT, which suggests the robustness of our infinite polymer sequence modeling approach.

Effectiveness of Backbone Embedding. We further assess the effectiveness of backbone embedding (BE). Specifically, we compare GIN3-512 with and without backbone embedding across eight polymer property prediction tasks equiped with the "star linking" strategy. The results are shown in Table 2. Model with BE outperforms the model without BE on Egc and Xc datasets, and is on par with the model without BE on the other 6 datasets. To understand these results, we calcuate the average number of rings per polymer and the ratio of polymers with more than two rings across each dataset, as shown in Figure 4. Notably, the Egc and Xc datasets exhibit the largest average number of rings per polymer and the largest ratio of polymers with more than two rings. Since backbone embedding is specifically designed to differentiate between the backbone structure and rings on the side chains, these ring statistics align with the observed model performance, highlighting the importance of backbone embedding for datasets with more ring structures.

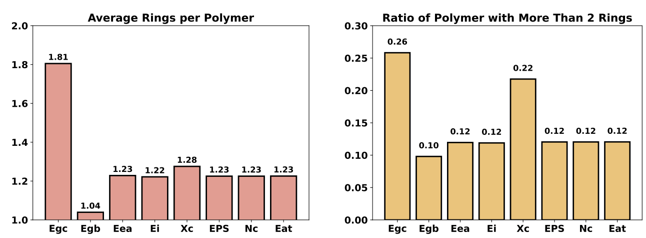

Figure 4: Ring statistics of eight polymer property datasets. Left: Average number of rings per polymer. Right: Ratio of polymers with more than two rings.

4.2 Multimodal Pre-training and Fine-tuning

Hyperparameters. For pre-training, we use $L = 6$ layers for the localized graph transformer and $d = 512$ hidden units. The atom mask rate p_{mask} is set to 0.3. We optimize the model using Adam optimizer [15] with $(\beta_1, \beta_2) = (0.9, 0.999)$, a learning rate of $2e-4$, and a batch size of 1024. We pre-train the model for 20,000 steps with 2,000 warmup steps. In fine-tuning stage, the model is trained for 50 epochs with a learning rate of $3e-5$ and a batch size of 32. All experiments are conducted on a single NVIDIA 4090 GPU.

Baseline Methods. We compare our method against several state-of-the-art methods, including four molecular pre-training methods: ChemBERTa [2], MolCLR [47], 3D Infomax [40], Uni-Mol [57], as well as five polymer pre-training methods SML [56], PLM [56], polyBERT [18], Transpolymer [50], MMPolymer [44]. The baseline methods are set up following the default settings provided in their original references.

Results. The performance comparison of different methods across eight polymer property datasets is presented in Table 4. It is clear that our MIPS outperforms the baseline methods on all datasets by considerable margins, especially on EPS and Eea datasets. Note that

Table 5: 2D: only use 2D graph with star linking; 2D + BE: add backbone embedding; 3D: only use 3D descriptors with a prediction head; 2D + BE + 3D: add backbone embedding and 3D descriptors. R^2 is reported for each dataset.

	Egc	Egb	Eea	Ei	Xc	EPS	Nc	Eat
2D	$0.904_{\pm 0.004}$	$0.932_{\pm 0.007}$	$0.931_{\pm 0.024}$	$0.835_{\pm 0.044}$	$0.409_{\pm 0.047}$	$0.803_{\pm 0.070}$	$0.870_{\pm 0.054}$	$0.980_{\pm 0.011}$
2D + BE	$0.915_{\pm 0.007}$	$0.934_{\pm 0.011}$	$0.933_{\pm 0.016}$	$0.836_{\pm 0.056}$	$0.457_{\pm 0.092}$	$0.806_{\pm 0.065}$	$0.873_{\pm 0.054}$	$0.988_{\pm 0.003}$
3D	$0.886_{\pm 0.010}$	$0.904_{\pm 0.018}$	$0.868_{\pm 0.027}$	$0.778_{\pm 0.052}$	$0.419_{\pm 0.044}$	$0.725_{\pm 0.051}$	$0.822_{\pm 0.031}$	$0.969_{\pm 0.015}$
2D + BE + 3D	$\mathbf{0.926}_{\pm 0.006}$	$\mathbf{0.945}_{\pm 0.007}$	$\mathbf{0.940}_{\pm 0.018}$	$\mathbf{0.846}_{\pm 0.051}$	$\mathbf{0.506}_{\pm 0.074}$	$\mathbf{0.814}_{\pm 0.045}$	$\mathbf{0.877}_{\pm 0.046}$	$\mathbf{0.990}_{\pm 0.004}$

MMPolymer [44] also adopts the a multimodal pre-training strategy that integrates 1D P-SMILES and 3D geometric information. However, our MIPS framework stands out by directly modeling infinite polymer sequences, achieving superior performance.

5 Ablation and Interpretation Study

In this section, we perform an ablation study to assess the contribution of various components in our model and explore the fragment-level interpretability. Specifically, we evaluate the efficacy of the backbone embedding and the structure encoder.

Effectiveness of backbone embedding and 3D encoder. We evaluate the impact of backbone embedding and the 3D encoder by comparing model performance with and without these components, as shown in Table 5. Comparing the first two rows, we observe that incorporating backbone embedding leads to improved performance, particularly on the Egc and Xc datasets, which aligns with the observations in Table 2. Comparing the second and fourth rows, it is evident that the inclusion of the 3D encoder enhances performance across all datasets. Furthermore, analyzing the last three rows reveals that combining both 2D and 3D encoders delivers the best overall performance, illustrating the synergistic role of the cross-modal fusion mechanism in MIPS.

5.1 Model Interpretation by Fragment Analysis

We employ fragment analysis to interpret the model, identifying the relative importance of different fragments across each dataset. Principal subgraph mining with 1-overlapping degree [17, 45] with 100 vocabulary size is used as the fragmentation method. The fragment library is constructed by the training set of each dataset. Additionally, we extend the fragment-level class activation mapping method [45] to regression tasks, which determines the relative importance of fragments specific to each dataset. For visualization purposes, we illustrate the structure and contribution scores of the three fragments with the highest importance and the three with the lowest importance. Results are presented in Figure 5. We further analyze the consistency of fragment importance with chemical knowledge. we choose Egc dataset as an example.

Egc. The objective of this task is to predict the bandgap of polymer chains. Fragments such as "COC", "CON", and "CCCCC" show higher-than-average contribution scores to the bandgap (chain), whereas fragments like "c1ccsc1", "C#C", and "C=S" exhibit lower-than-average scores. These findings align with chemical knowledge. COC and CON typically exhibit localized electron density due to the electronegative oxygen or nitrogen atoms. They stabilize the molecule but do not participate strongly in π-conjugation [37]. Consequently, their presence slightly increases the band gap. "CCCCC" (Linear Alkyl Chain) is nonpolar and non-conjugated [13]. It does not contribute to the π-electron delocalization directly and slightly increases the band gap by reducing molecular excitability. On the other hand, "c1ccsc1", "C#C, "C=S" actively contribute to π-conjugation [25], which often leads to a reduction in the bandgap.

From the analysis above, we conclude that the model can learn meaningful representations of polymers, which reflect the contributions of different functional groups.

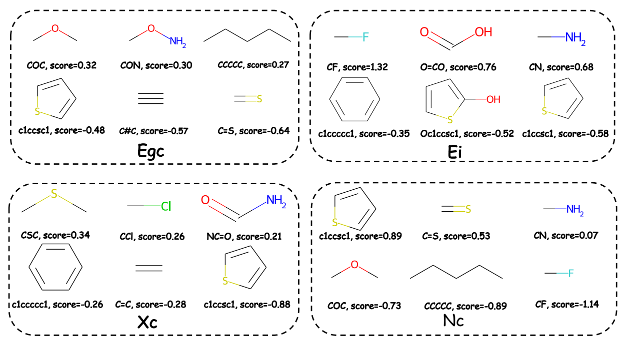

Figure 5: Visualization of the top three and bottom three fragments with the highest and lowest importance scores on Egc, Ei, Xc, and Nc dataset.

6 Conclusion

In this work, we introduce the Multimodal Infinite Polymer Sequence (MIPS) pre-training framework for polymer property prediction. Specifically, we propose a novel "star linking" strategy and backbone embedding to generalize the message passing and graph attention mechanisms to infinite polymer sequences with enhanced capability beyond Weisfeiler-Lehman test, which is provably robust under repeat and shift invariance test. Additionally, we design a cross-modal module to effectively fuse the topological and spatial information, enabling the model to learn multimodal representations via masked-atom prediction. Extensive experimental results across eight polymer property prediction tasks verify the efficacy of our method, which achieves the state-of-the-art performance. Ablation studies underscore the utility of the proposed modules, while the fragment analysis further provides evidence of the model's ability to capture chemically meaningful polymer representations.

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